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University ST4005CEM Database System

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1. Find the total billing amount paid by each patient. Display PatientID and total amount.

```
173
174 • select
175     patientID, sum(amount) as total
176 from billing group by patientID;
177
```

Result Grid |  Filter Rows: Export:

	patientID	total
▶	2	1500.00
	3	2300.50
	5	3700.00
	6	4750.25
	101	9100.00

2. Count the total number of appointments booked for each doctor. Display Doctorname and appointment count.

```
178 • select
179     concat(doctor.firstName," ",doctor.lastName) as Doctors,
180     count(appointment.appointmentID) as appointments
181 from doctor
182 join appointment
183 on doctor.doctorID=appointment.doctorID group by doctor.doctorID;
184
```

Result Grid |  Filter Rows: Export:  Wrap Cell Content: 

	Doctors	appointments
▶	Dr. raman khatiwada	1
	Rajesh Karki	2
	Suresh Pandey	1
	Mina Tamang	1
	Dipak Adhikari	1

3. Find the average billing amount from the Billing table.

```
184
185 • select avg(amount)
186     as avg_amount
187     from billing;
188
```

Result Grid |   Filter Rows:

	avg_amount
▶	2668.843750

4. Display the maximum and minimum billing amount recorded in the Billing table.

```
189 • select max(amount)
190     as max_bill_amount
191     from billing;
192
```

Result Grid |   Filter Rows:

	max_bill_amount
▶	5000.00

```
---
189 • select min(amount)
190     as max_bill_amount
191     from billing;
192
```

Result Grid |   Filter Rows:

	max_bill_amount
▶	500.00

5. List all doctors who have issued more than 1 prescriptions. Display doctorname and prescription count.

```
193 • select
194     concat(d.firstName," ",d.lastName) as Doctors,
195     count(p.prescriptionID)
196 as prescription_count
197 from prescription p join doctor d on d.doctorID=p.doctorID
198 group by p.doctorID having count(p.prescriptionID) >1 ;
199
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	Doctors	prescription_count
▶	Rajesh Karki	2

6. Find the total number of medical records grouped by Diagnosis. Display Diagnosis and record count.

```
212
213 • select diagnosis,count(diagnosis) as record_count
214 from medicalRecord
215 group by (diagnosis);
216
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell C

	diagnosis	record_count
▶	Hypertension	2
	Diabetes	2
	Asthma	1
	Flu	1

7. Display the full name of each patient along with their diagnosis from the MedicalRecord table.

```
218 • select concat(p.firstName, " ", p.lastName) as patientName,
219       m.diagnosis from medicalRecord m
220       join patient p on p.patientID=m.patientID;
221
222
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	patientName	diagnosis
▶	Ram Sharma	Hypertension
	Priya Thapa	Diabetes
	Bikash Rai	Hypertension
	Sita Gurung	Asthma
	Rohan Magar	Diabetes
	Aarav Sharma	Flu

8. List all appointments along with the patient's first name, last name, and the doctor's first name and last name.

```
225 • select
226       a.appointmentID, concat(p.firstName, " ", p.lastName) as PatientName,
227       concat(d.firstName, " ", d.lastName) as DoctorName, a.reason, a.status
228 from appointment a join patient p on p.patientID=a.patientID
229 join doctor d on d.doctorID=a.doctorID ;
230
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	appointmentID	PatientName	DoctorName	reason	status
▶	1	Ram Sharma	Dr. raman khatiwada	All good	COMPLETED
	2	Ram Sharma	Rajesh Karki	Chest pain checkup	Completed
	11	Rohan Magar	Rajesh Karki	Heart rate monitoring	Scheduled
	4	Bikash Rai	Suresh Pandey	Knee joint pain	Completed
	5	Sita Gurung	Mina Tamang	Child fever	Completed
	6	Rohan Magar	Dipak Adhikari	Skin rash treatment	Scheduled

9. Show all prescriptions along with the medication name, patient name, and the prescribing doctor's name.

```

232 • select concat(p.prescriptionID," ", p.medication," ",p.dosage," ",p.startDate," ",p.endDate )
233      as prescriptions , concat(pt.firstName, " ",pt.lastName) as patientName,
234      concat(d.firstName, " ",d.lastName) as DoctorName
235 from prescription p join patient pt on pt.patientID= p.patientID
236 join doctor d on d.doctorID=d.doctorID ;
237

```

	prescriptions	patientName	DoctorName
▶	13 Amoxicillin 500mg twice 2025-01-01 2025-02-10	Ram Sharma	Dr. raman khatiwada
	18 Paracetamol 650mg thrice 2025-02-01 2025-03-12	Aarav Sharma	Dr. raman khatiwada
	17 Cetirizine 10mg once 2025-01-20 2025-03-01	Rohan Magar	Dr. raman khatiwada
	16 Atorvastatin 20mg once 2025-01-15 2025-03-16	Sita Gurung	Dr. raman khatiwada
	15 Metformin 850mg twice 2025-01-10 2025-03-21	Bikash Rai	Dr. raman khatiwada
	20 Painkiller 50 mg twice 2026-05-13 2026-06-02	Priya Thapa	Dr. raman khatiwada
	19 Painkiller 50 mg twice 2026-05-13 2026-06-02	Priya Thapa	Dr. raman khatiwada
	14 Ibuprofen 400mg thrice 2025-01-05 2025-02-14	Priya Thapa	Dr. raman khatiwada
	13 Amoxicillin 500mg twice 2025-01-01 2025-02-10	Ram Sharma	Rajesh Karki
	18 Paracetamol 650mg thrice 2025-02-01 2025-03-12	Aarav Sharma	Rajesh Karki
	17 Cetirizine 10mg once 2025-01-20 2025-03-01	Rohan Magar	Rajesh Karki
	16 Atorvastatin 20mg once 2025-01-15 2025-03-16	Sita Gurung	Rajesh Karki
	15 Metformin 850mg twice 2025-01-10 2025-03-21	Bikash Rai	Rajesh Karki

10. List all patients and their billing details. Include patients who have no billing records as well.

```

240 • select * from patient
241 left join billing on
242 patient.patientID=billing.patientID;
243

```

	patientID	firstName	lastName	dateOfBirth	gender	Phone	BloodGroup	billingID	patientID	amount	billindDate	status
▶	2	Priya	Thapa	1985-07-22	female	9812345678	NULL	2	2	1500.00	2025-01-11	Completed
	3	Bikash	Rai	2000-11-08	male	9823456789	NULL	3	3	2300.50	2025-01-16	Completed
	4	Sita	Gurung	1995-01-30	female	9834567890	NULL	NULL	NULL	NULL	NULL	NULL
	5	Rohan	Magar	1978-06-12	others	9845678901	NULL	5	5	3200.00	2025-02-02	Completed
	5	Rohan	Magar	1978-06-12	others	9845678901	NULL	10	5	500.00	2025-03-16	Pending
	6	Aarav	Sharma	1990-03-15	male	9801234567	NULL	6	6	950.25	2025-02-11	Pending
	6	Aarav	Sharma	1990-03-15	male	9801234567	NULL	11	6	3800.00	2025-03-21	Pending
	101	Ram	Sharma	1990-05-10	male	9812345678	NULL	1	101	5000.00	2026-05-08	Pending
	101	Ram	Sharma	1990-05-10	male	9812345678	NULL	7	101	4100.00	2025-02-21	Completed

11. Display all doctors and any appointments they have. Show doctors with no appointments too.

```
247 • select * from doctor d left join appointment a on d.doctorID=a.doctorID;
248
```

	doctorID	firstName	lastName	speciality	phone	email	appointmentID	patientID	doctorID	appointmentDate	reason
▶	1	Dr. raman	khatiwada	Cardiology	9812311222	ramankhatiwada@hospital.com	1	101	1	2026-05-08	All good
	2	Rajesh	Karki	Cardiology	9851234567	rajesh@gmail.com	2	101	2	2025-01-10	Chest pain chec
	2	Rajesh	Karki	Cardiology	9851234567	rajesh@gmail.com	11	5	2	2025-03-20	Heart rate moni
	3	Anita	Shrestha	Neurology	9862345678	anita@gmail.com	NULL	NULL	NULL	NULL	NULL
	4	Suresh	Pandey	Orthopedics	9873456789	suresh@gmail.com	4	3	4	2025-01-20	Knee joint pain
	5	Mina	Tamang	Pediatrics	9884567890	mina@gmail.com	5	4	5	2025-02-01	Child fever
	6	Dipak	Adhikari	Dermatology	9895678901	dipak@gmail.com	6	5	6	2025-02-10	Skin rash treatm

12. Find each patient's full name and the total amount billed to them by joining Patient and Billing tables.

```
249 • select concat(p.firstname," ",p.lastName) as Patient_name,
250 sum(b.amount) as Total_bill
251 from patient p join billing b on p.patientID=b.patientID
252 group by (p.patientID) ;
253
```

	Patient_name	Total_bill
▶	Priya Thapa	1500.00
	Bikash Rai	2300.50
	Rohan Magar	3700.00
	Aarav Sharma	4750.25
	Ram Sharma	9100.00

13. List appointment details (date, reason, status) together with the patient's gender and contact info.

```
254 • select concat( a.appointmentdate,a.reason,a.status)
255 as appointment, p.gender as gender,p.phone as contact
256 from appointment a join patient p on a.patientID=p.patientID;
257
```

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
appointment	gender	contact	
2026-05-08All goodCOMPLETED	male	9812345678	
2025-01-10Chest pain checkupCompleted	male	9812345678	
2025-01-20Knee joint painCompleted	male	9823456789	
2025-02-01Child feverCompleted	female	9834567890	
2025-02-10Skin rash treatmentScheduled	others	9845678901	
2025-03-20Heart rate monitoringScheduled	others	9845678901	

14. Retrieve all appointments where the status is 'Scheduled' and appointments is after Jan 1 2026.

```
259 • select * from appointment
260 where appointmentDate > "2026-01-01" and status ="Scheduled";
261
```

Result Grid						
Filter Rows:		Edit:	Export/Import:			
appointmentID	patientID	doctorID	appointmentDate	reason	status	
*	NULL	NULL	NULL	NULL	NULL	

15. Find all patients whose first name starts with the letter 'A'.

```
262 • select * from patient where firstName like "a%";
```

```
263
```

Result Grid							
		Filter Rows:		Edit:		Export/Import:	
	patientID	firstName	lastName	dateOfBirth	gender	Phone	BloodGroup
▶	6	Aarav	Sharma	1990-03-15	male	9801234567	NULL
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

16. Display all billing records where the amount is between 500 and 2000.

```
263
```

```
264 • select * from billing where amount between 500 and 2000;
```

```
265
```

Result Grid					
		Filter Rows:		Edit:	
	billingID	patientID	amount	billindDate	status
▶	2	2	1500.00	2025-01-11	Completed
	6	6	950.25	2025-02-11	Pending
	10	5	500.00	2025-03-16	Pending
*	NULL	NULL	NULL	NULL	NULL

17. List all prescriptions where the medication is either 'Paracetamol', 'Ibuprofen', or 'Amoxicillin'.

```
265
266 • select * from prescription
267 |where medication in ( 'Paracetamol', 'Ibuprofen', 'Amoxicillin' ) ;
268
```

Result Grid



 Filter Rows:

Edit:   

Export/Import:  

Wra

	prescriptionID	patientID	doctorID	medication	dosage	startDate	endDate
▶	13	101	1	Amoxicillin	500mg twice	2025-01-01	2025-02-10
	14	2	2	Ibuprofen	400mg thrice	2025-01-05	2025-02-14
	18	6	3	Paracetamol	650mg thrice	2025-02-01	2025-03-12
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

18. Retrieve medical records where the treatment contains the word 'Surgery' or the notes are not null.

```
269 • SELECT * FROM medicalrecord
270 WHERE treatment LIKE "%surgery%" OR treatment IS NOT NULL;
271 • select * from patient;
```

Result Grid

  Filter Rows:

Edit:   

Export/Import:  

	recordID	patientID	diagnosis	treatment	recordDate	Notes
▶	1	101	Hypertension	Medication	2026-05-08	Follow the instruction
	9	2	Diabetes	Insulin Therapy	2026-05-13	Monitor blood sugar daily
	10	3	Hypertension	Medication	2026-05-13	Reduce salt intake
	11	4	Asthma	Inhaler	2026-05-13	Avoid dust and smoke
	12	5	Diabetes	Oral Medication	2026-05-13	Check HbA1c monthly
	13	6	Flu	Rest and Fluids	2026-05-13	Stay hydrated
*	NULL	NULL	NULL	NULL	NULL	NULL

19. Add a new column called BloodGroup of type VARCHAR(5) to the Patient table.

272

```
273 • desc patient;
```

274

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	patientID	int	NO	PRI	NULL	
	firstName	varchar(20)	YES		NULL	
	lastName	varchar(20)	YES		NULL	
	dateOfBirth	date	YES		NULL	
	gender	enum('male','female','others')	YES		NULL	
	Phone	varchar(15)	YES		NULL	
	BloodGroup	varchar(5)	YES		NULL	

20. Modify the Phone column in the Doctor table to change its data type from VARCHAR(15) to VARCHAR(20).

```
278 • ALTER TABLE doctor MODIFY COLUMN phone VARCHAR(20);
```

```
279 • DESC doctor;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	doctorID	int	NO	PRI	NULL	
	firstName	varchar(20)	YES		NULL	
	lastName	varchar(20)	YES		NULL	
	speciality	varchar(20)	YES		NULL	
	phone	varchar(20)	YES		NULL	
	email	varchar(50)	YES		NULL	